

Jialong Wu

wujialong0229@gmail.com | manchery.github.io | [Google Scholar](#)

EDUCATION

Ph.D. in Software Engineering School of Software, Tsinghua University • Advisor: Prof. Mingsheng Long • Research Interests: World Models, Reinforcement Learning	Aug 2022 – June 2027 (<i>expected</i>) Beijing, China
Bachelor in Software Engineering School of Software, Tsinghua University • GPA: 3.98/4.0, Rank: 1/89	Aug 2018 – June 2022 Beijing, China
Bachelor in Pure and Applied Mathematics (Second Degree) Department of Mathematical Sciences, Tsinghua University • GPA: 3.75/4.0	Aug 2019 – June 2022 Beijing, China

EXPERIENCE

Co-founder & Chief Research Officer ByteDance Seed	August 2026 – <i>present</i> Beijing, China
Research Intern ByteDance Seed	March 2025 – June 2026 Beijing, China

PUBLICATIONS AND PREPRINTS

Seed2.0 Model Card: Towards Intelligence Frontier for Real-World Complexity

Bytedance Seed: ... , **Jialong Wu** (Contributor), ... *Technical report*, 2026.
[\[Project website\]](#) [\[PDF\]](#)

TimeVista: Exploring and Exploiting Vision-Language Models as Judges for Time Series Forecasting

Zhi Chen*, Yuxuan Wang*, **Jialong Wu**#, Yong Liu, Haoran Zhang, Xingjian Su, Jianmin Wang, Mingsheng Long#. *arXiv preprint*, 2026.
[\[PDF\]](#)

OpenWorldLib: A Unified Codebase and Definition of Advanced World Models

DataFlow Team, ... , **Jialong Wu** (Contributor), ... *Technical report*, 2026.
[\[Project website\]](#) [\[PDF\]](#) [\[Code\]](#)

Seed1.8 Model Card: Towards Generalized Real-World Agency

Bytedance Seed: ... , **Jialong Wu** (Contributor), ... *Technical report*, 2026.
[\[Project website\]](#) [\[PDF\]](#)

Visual Generation Unlocks Human-Like Reasoning through Multimodal World Models

Jialong Wu, Xiaoying Zhang#, Hongyi Yuan, Xiangcheng Zhang, Tianhao Huang, Changjing He, Chaoyi Deng, Renrui Zhang, Youbin Wu, Mingsheng Long#. *arXiv preprint*, 2026.
[\[Project website\]](#) [\[PDF\]](#) [\[Code\]](#) [\[Slides\]](#) [\[Datasets\]](#)

Thoth: Mid-Training Bridges LLMs to Time Series Understanding

Jiafeng Lin*, Yuxuan Wang*, **Jialong Wu***, Huakun Luo, Zhongyi Pei#, Jianmin Wang. *arXiv preprint*, 2026.
[\[PDF\]](#) [\[Code\]](#) [\[Models\]](#)

JEPA-VLA: Video Predictive Embedding is Needed for VLA Models

Shangchen Miao, Ningya Feng, **Jialong Wu**, Ye Lin, Xu He, Dong Li, Mingsheng Long#. *arXiv preprint*, 2026.
[\[PDF\]](#)

Audio-Visual World Models: Towards Multisensory Imagination in Sight and Sound

Jiahua Wang*, Shannan Yan*, Leqi Zheng*, **Jialong Wu**, Yaixin Mao. *arXiv preprint*, 2025.
[\[PDF\]](#)

Vid2World: Crafting Video Diffusion Models to Interactive World Models

Siqiao Huang*, **Jialong Wu***, Qixing Zhou, Shangchen Miao, Mingsheng Long#. *ICLR*, 2026.

[\[Project website\]](#) [\[PDF\]](#) [\[Code\]](#)

Foresight Diffusion: Improving Sampling Consistency in Predictive Diffusion Models

Yu Zhang*, Xingzhuo Guo*, Haoran Xu, **Jialong Wu**, Mingsheng Long#. *ICLR*, 2026.

[\[PDF\]](#)

RLVR-World: Training World Models with Reinforcement Learning

Jialong Wu, Shaofeng Yin, Ningya Feng, Mingsheng Long#. *NeurIPS*, 2025.

[\[Project website\]](#) [\[PDF\]](#) [\[Code\]](#) [\[Models & Datasets\]](#) [\[Poster\]](#)

Seed1.5-VL Technical Report

Dong Guo, Faming Wu, Feida Zhu, Fuxing Leng, Guang Shi, ... , **Jialong Wu** (Contributor), ... *Technical report*, 2025.

[\[Project website\]](#) [\[PDF\]](#) [\[Code\]](#)

Trajectory World Models for Heterogeneous Environments

Shaofeng Yin*, **Jialong Wu***, Siqiao Huang, Xingjian Su, Xu He, Jianye Hao, Mingsheng Long#. *ICML*, 2025.

[\[PDF\]](#) [\[Code\]](#)

CompilerDream: Learning a Compiler World Model for General Code Optimization

Chaoyi Deng*, **Jialong Wu***, Ningya Feng, Jianmin Wang, Mingsheng Long#. *KDD*, 2025.

[\[PDF\]](#) [\[Code\]](#)

Long-Sequence Recommendation Models Need Decoupled Embeddings

Ningya Feng*, Junwei Pan*, **Jialong Wu***, Baixu Chen, Ximei Wang,

Qian Li, Xian Hu, Jie Jiang, Mingsheng Long#. *ICLR*, 2025.

[\[PDF\]](#) [\[Code\]](#)

iVideoGPT: Interactive VideoGPTs are Scalable World Models

Jialong Wu*, Shaofeng Yin*, Ningya Feng, Xu He, Dong Li, Jianye Hao, Mingsheng Long#. *NeurIPS*, 2024.

[\[Project Website\]](#) [\[PDF\]](#) [\[Code\]](#) [\[Models\]](#) [\[Poster\]](#) [\[Slides\]](#) [\[Blog\]](#)

HarmonyDream: Task Harmonization Inside World Models

Haoyu Ma*, **Jialong Wu***, Ningya Feng, Chenjun Xiao, Dong Li, Jianye Hao, Jianmin Wang, Mingsheng Long#. *ICML*, 2024.

[\[PDF\]](#) [\[Code\]](#) [\[Poster\]](#) [\[Slides\]](#) [\[Talk\]](#) [\[Blog\]](#)

Pre-training Contextualized World Models with In-the-wild Videos for Reinforcement Learning

Jialong Wu*, Haoyu Ma*, Chaoyi Deng, Mingsheng Long#. *NeurIPS*, 2023.

[\[PDF\]](#) [\[Code\]](#) [\[Poster\]](#) [\[Slides\]](#) [\[Blog\]](#)

CLIPood: Generalizing CLIP to Out-of-Distributions

Yang Shu*, Xingzhuo Guo*, **Jialong Wu**, Ximei Wang, Jianmin Wang, Mingsheng Long#. *ICML*, 2023.

[\[PDF\]](#) [\[Code\]](#)

Supported Policy Optimization for Offline Reinforcement Learning

Jialong Wu, Haixu Wu, Zihan Qiu, Jianmin Wang, Mingsheng Long#. *NeurIPS*, 2022.

[\[PDF\]](#) [\[Code\]](#) [\[Poster\]](#) [\[Slides\]](#)

Out-of-Dynamics Imitation Learning from Multimodal Demonstrations

Yiwen Qiu, **Jialong Wu**, Zhangjie Cao, Mingsheng Long#. *CoRL*, 2022.

[\[PDF\]](#) [\[Code\]](#)

Flowformer: Linearizing Transformers with Conservation Flows

Haixu Wu, **Jialong Wu**, Jiehui Xu, Jianmin Wang, Mingsheng Long#. *ICML*, 2022.

[\[PDF\]](#) [\[Code\]](#)

BLOGS AND POSITIONS

Building Physical RSI Beyond the Known World

MirroS Team, *August* 2026.

[\[Blog\]](#) [\[Video\]](#)

HONORS AND AWARDS

ICML 2026 Gold Reviewer ICML 2026 [Link]	2026
ByteDance Scholarship (awarded to 20 students in China and Singapore) ByteDance [Certificate]	2025
ICLR 2025 Notable Reviewer ICLR 2025 [Link]	2025
National Scholarship (Top 2%) Ministry of Education of the People's Republic of China [Certificate 2020] [Certificate 2024]	2020, 2024
Excellent Teaching Assistant Award Tsinghua University [Certificate]	2024
NeurIPS 2023 Top Reviewer NeurIPS 2023 [Link] [Certificate]	2023
Future Researcher Scholarship Tsinghua University [Certificate]	2022
Outstanding Graduates (Top 2%) School of Software [Certificate] & Tsinghua University [Certificate] & Beijing [Certificate]	2022
Excellent Graduation Thesis Tsinghua University [Certificate]	2022
Comprehensive Excellence Award (Top 10%) Tsinghua University [Certificate 2019] [Certificate 2021] [Certificate 2023]	2019, 2021, 2023, 2025
34th National Olympiad in Informatics (NOI 2017), Silver Medal (81st Place) China Computer Federation (CCF) [Certificate] [Link]	July 2017

INVITED TALKS AND ACTIVITIES

Bridging World Models and Multimodal Reasoning MAR 2026 workshop with CVPR [Slides]	June, 2026
On Concepts of World Models: From Simulation to Thinking NICE AI Talk [Slides]	April, 2026
World Models and Human-like Reasoning MERL Seminar Series	March, 2026
World Models and Human-like Reasoning Google DeepMind London	Feb, 2026
A Path Towards General World Models BAAI Conference, Beijing, China	June, 2025
Panelist AIFC Side Event of the AI Action Summit 2025 , Paris, France	Feb, 2025
Towards General World Models: Pre-training, Multi-Modality, and Scalable Architecture ByteDance Research	June, 2024
Intrinsic Task Harmonization Inside World Models RLChina	Feb, 2024

PROFESSIONAL SERVICES

Reviewer | PC Member

- International Conference on Learning Representations (*ICLR*) 2024 – 2026
- Conference on Neural Information Processing Systems (*NeurIPS*) 2023 – 2025
- International Conference on Machine Learning (*ICML*) 2022 – 2026
- Conference on Computer Vision and Pattern Recognition (*CVPR*) 2026
- Transactions on Machine Learning Research (*TMLR*)

Teaching Assistant

- Deep Learning, Fall 2023, Tsinghua University. Instructor: Prof. Mingsheng Long.
- Introduction to Algorithms, Spring 2023, Tsinghua University. Instructor: Prof. Bin Wang.
- Machine Learning, Fall 2022, Tsinghua University. Instructor: Prof. Mingsheng Long.
- Introduction to Artificial Intelligence, Spring 2021, 2022, 2024, Tsinghua University. Instructor: Prof. Mingsheng Long.

Student Counselor

- Undergraduate Student Counselor, Dushi College, Tsinghua University, 2024–2027 (*expected*).

SELECTED PROJECTS

MiniVeo3-Reasoner: Thinking with Videos from Open-Source Priors | 230+ Stars 2025

We reproduce chain-of-frames visual reasoning by fine-tuning open-source video models.

Awesome Multi-task Learning | 840+ Stars 2021 – 2025

A curated list of datasets, codebases, and papers on Multi-Task Learning (MTL).

ImageLabeler | C++, Qt | 120+ Stars, 40+ Forks Summer 2019

Image annotation tool, supporting 2D/3D detection/segmentation annotation.

Incremental Decision Tree | Python (NumPy, Matplotlib, Pandas) Fall 2020

Re-implementation of VFDT and EFDT, reproducing paper *Extremely Fast Decision Tree*.

TECHNICAL SKILLS

Programming Language: Python, C/C++, MATLAB, Java, HTML/CSS, JavaScript

Professional Software: PyTorch, JAX, NumPy, Git, LaTeX, Vue

Language: China (native), English

Last updated: August 7, 2026